IT3021 – Data Warehousing & Business Intelligence
Year 3 Semester 1 & 2
Lab Sheet 07

1. Introduction to the Worksheet

In this worksheet, we will create an OLAP cube using SQL Server Analysis Services; SSAS. In order to continue with this worksheet, all SSIS practical worksheets (worksheet 1 through 5) must be completed and data must be available in the data warehouse layer.

2. Tools Required

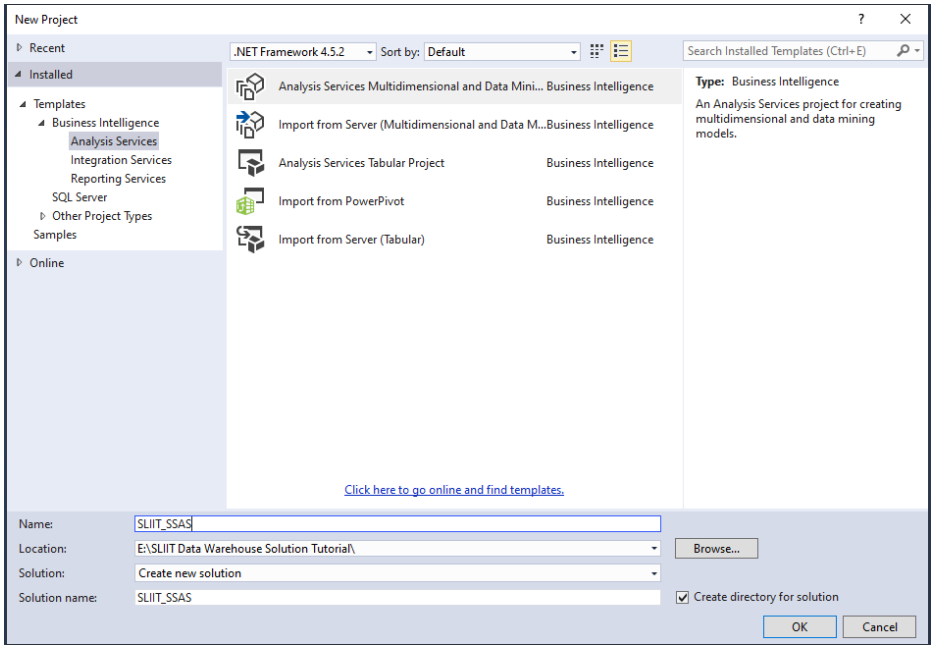
Below tools should be available to continue with this tutorial:

- SQL Server Data Tools
- SQL Server Management Studio

3. Creating the SSAS Project

In order to create required Cubes and related components, let's create an SSAS Project using Visual Studio Data Tools.

1. Open Visual Studio Data Tools in 'Administrator' mode.
2. Create an ***Analysis Services*** project.
    a. From the toolbar, click on ***File → New → Project***



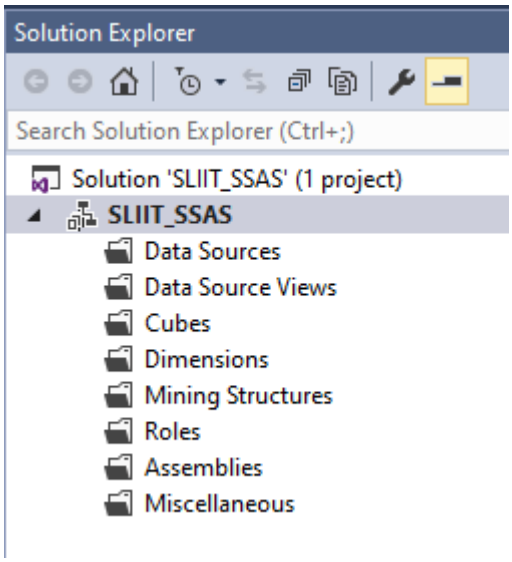
b. In the ***New Project*** window, under ***Template*** section, expand ***Business Intelligence***, select ***Analysis Services***

c. From the available templates, select ***Analysis Services Multidimensional and Data Mining Project***.

d. Provide a name; '**SLIIT_SSAS**', and a location**;** '**C:\SLIIT Data Warehouse Solution Tutorial**' and click ***OK***.

e. Optionally you can select the ***Create directory for solution*** checkbox.

3. Once the solution is created, you should see a similar folder structure to the image below, in ***Solution Explorer***.



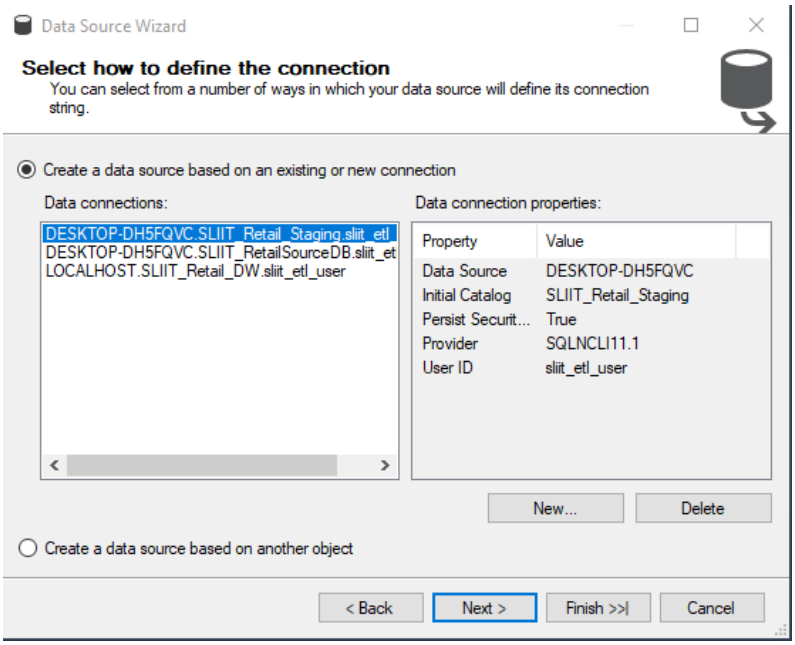
4. We need to configure components above, starting from ***Data Sources*** to ***Dimensions***, in order to create a working SSAS Cube.

4. Creating a Data Source

First step is to configure a data source. Data source defines from where, the cube is extracting data.

1. To configure, right click on ***Data Sources*** in and select ***New Data Source***…
2. In the ***Data Source Wizard***, click on the ***Next***.
3. In the ***Select how to define the connection*** window, select ***Create a data source based on an existing or new connection*** option.
4. You might be able to see a connection already to the data warehouse, '**SLIIT_Retail_DW**' as we have already created this connection previously in SSIS

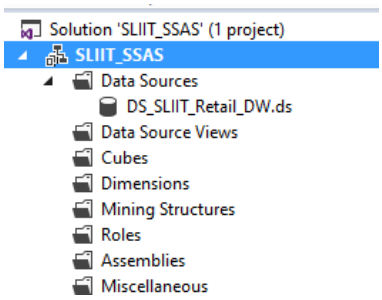
projects. If not, you can create a connection by clicking on ***New…*** button and following the onscreen instructions. Select the connection to '**SLIIT_Retail_DW**' and click ***Next***.



Note: The exact connection name, along with computer name may probably be different from what you see in the above image.

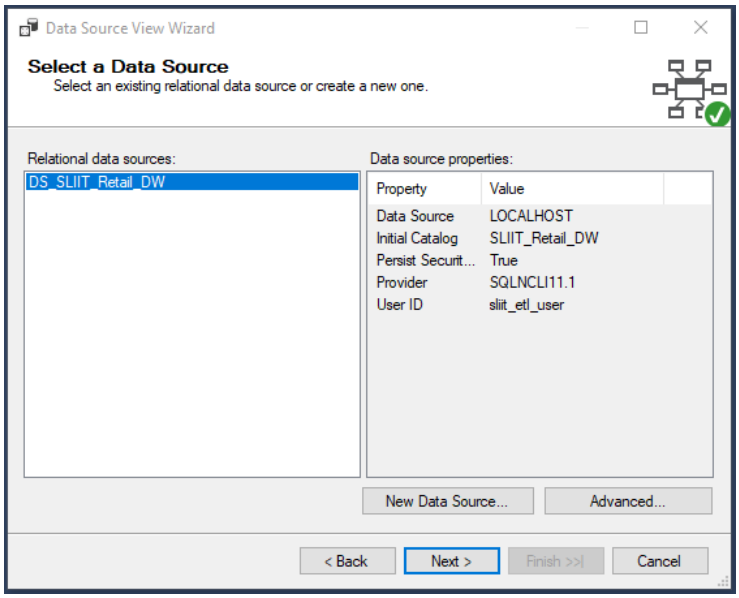
5. In the ***Impersonation Information*** page, select the method you use to connect to SQL Server Management Studio. Best option would be to use a service account. However, for this worksheet you can use any option. If you use the first option; ***Use a specific Windows user name and password***, you need to provide user name and password you use to login to Windows.

   If you provide invalid credentials, this step will not give you any warning or error. However, later you will get security errors. Thus ensure you provide correct credentials. Click ***Next***.
6. Provide a data source name; '**DS_SLIIT_Retail_DW**' and click ***Finish***.
7. You should be able to see the new data source you created under ***Data Sources***.

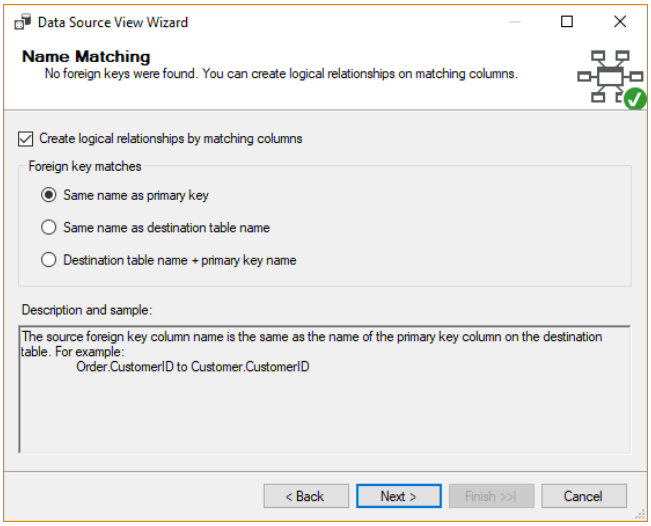


5. Creating a Data Source View

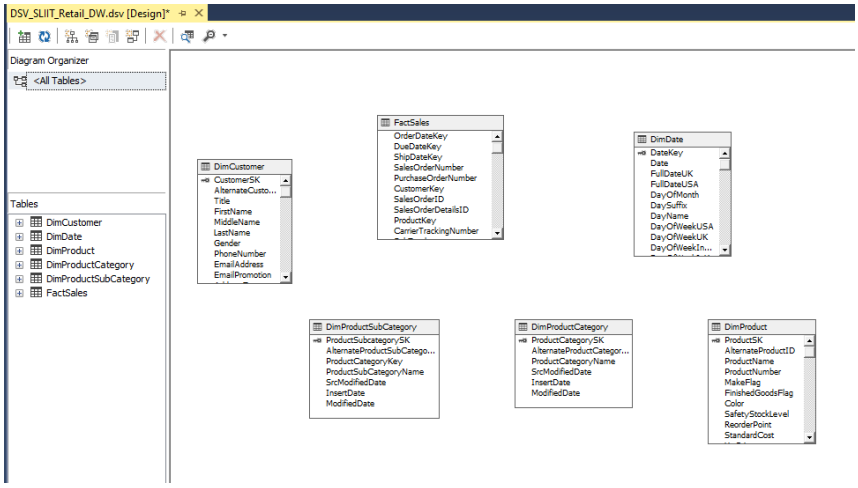
1. Right click on ***Data Source Views*** and select ***New Data Source View…***
2. In the ***Data Source View Wizard***, click on the ***Next***.
3. In the ***Select a Data Source*** page, select the data source you created in step 4 from the ***Relational data sources*** list and click ***Next***.



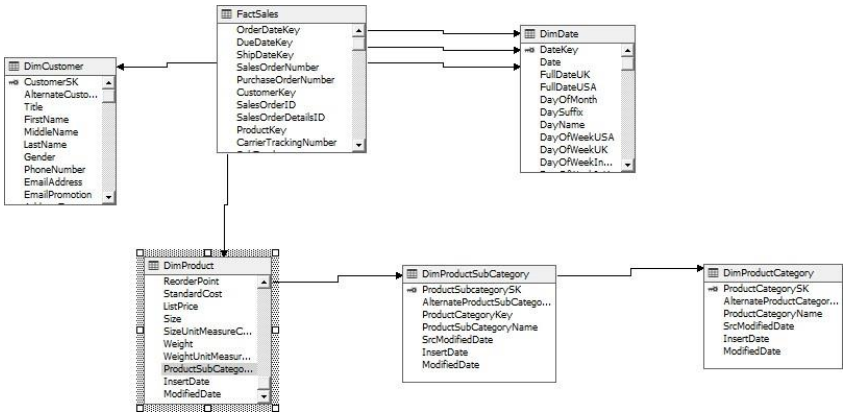
4. ***Name Matching*** page is about how to map dimension tables and measure (fact) tables automattically. Based on our design, primary key (surrogate key) name in dimension tables (which ends with '***SK***') is used in fact tables as in a different way (ends with '***Key***'). Thus, SSAS will not be able to map related tables automatically. If we had used the same column names in dimension tables and fact table foreign key columns, ***Same name as primary key*** option under ***Create logical relationships by matching columns*** would have generated mappings between tables automatically. Anyhow, let's select this option and move forward by clicking ***Next***.



5. In the ***Select Tables and Views*** page, first click on '***Fact Sales (dbo)***'and click on **>** button to move it to the ***Included objects*** window. Then click on ***Add Related Tables*** button. Note that nothing is getting selected automatically. This is because dimension tables' primary key field and fact table's foreign key columns are not having the same name. You can add all the tables to the ***Included objects*** window by clicking ***>>*** button. Click ***Next***.
6. Provide a data source view name; '**DSV_SLIIT_Retail_DW**' and click ***Finish***.
7. You will see that none of the tables are connected in the design view once completed.

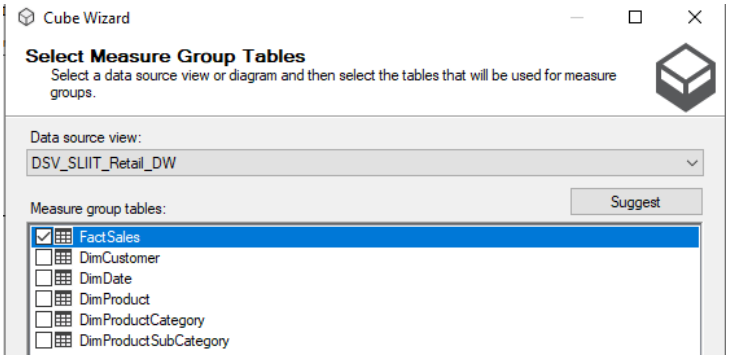


8. Create table links:
    a. To link '***DimCustomer***' and '***FactSales***' tables, click on '***CustomerKey***' column of the '***FactSales***', drag and drop it on the '***CustomerSK***' column of the '***DimCustomer***' table.
    b. Similarly using corresponding SK/FKs link, '**OrderDateKey**' of '**FactSales**' with '**DimDate**'
    c. '**DueDateKey**' of '**FactSales**' with '**DimDate**'
    d. '**ShipDateKey**' of '**FactSales**' with '**DimDate**'
    e. '**ProductKey**' of '**FactSales**' with '**DimProduct**'
    f. '**ProductSubCategoryKey**' of '**DimProduct**' with '**DimProductSubCategory**'
    g. '**ProductCategoryKey**' of '**DimProductSubCategory**' with '**DimProductCategory**'

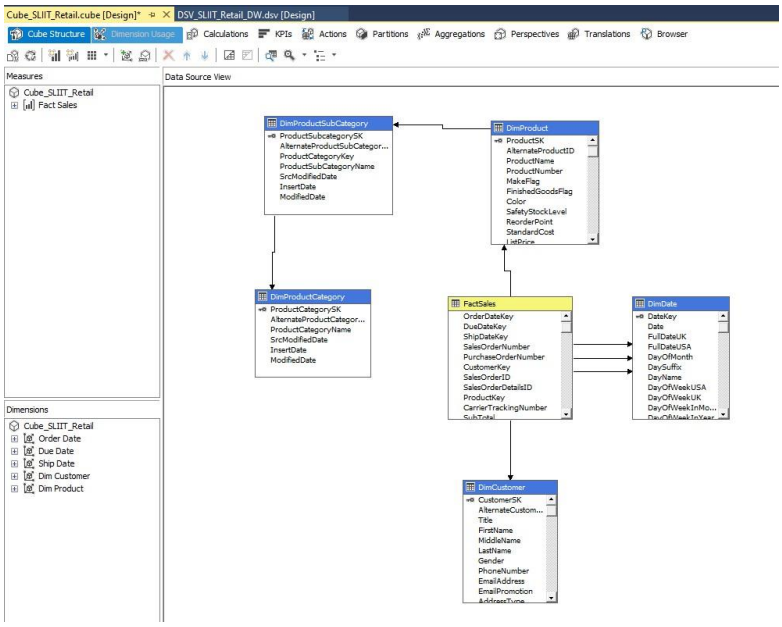


6. Creating a Cube

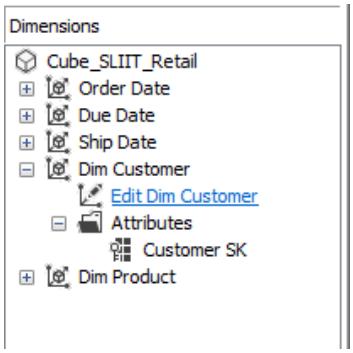
1. Right click on ***Cubes*** and select ***New Cube…***
2. In the ***Cube Wizard***, click on the ***Next***.
3. In the Select Creation Method page, select ***Use existing tables*** option and click ***Next***.
4. In the ***Select Measure Group Tables*** page, select the data source view created in step 5 (‘**DSV_SLIIT_Retail_DW**’) from the dropdown list. ***Suggest*** button will select probable fact tables based on the columns in the tables. Not that the wizard will suggest/select ‘**DimProduct**’ as a fact table. This is because ‘**DimProduct**’ table has measurable fields such as ‘**weight**’. However, only fact table is ‘**FactSales**’. Thus, select ‘**FactSales**’ and click ***Next***.



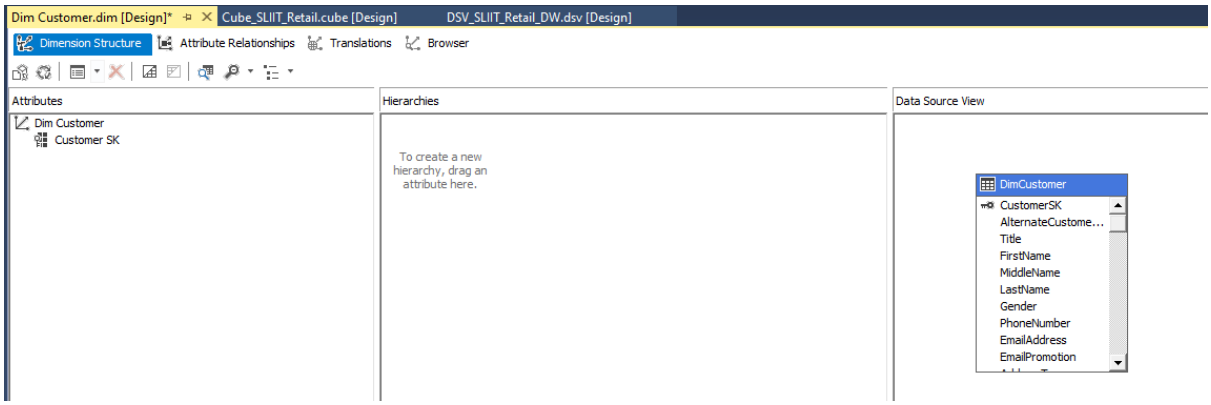
5. In the ***Select Measures*** page, select all the fields and click ***Next***. (Though fields such as “**Sales Order ID**” are not real measures, if needed we can use them for calculations such as count of sales)
6. In the ***Select New Dimensions*** page, select all the dimension tables and click ***Next***.
7. Provide a cube name; ‘**Cube_SLIIT_Retail**’ and click ***Finish***.
8. You will see the same snowflake schema is built as a cube.



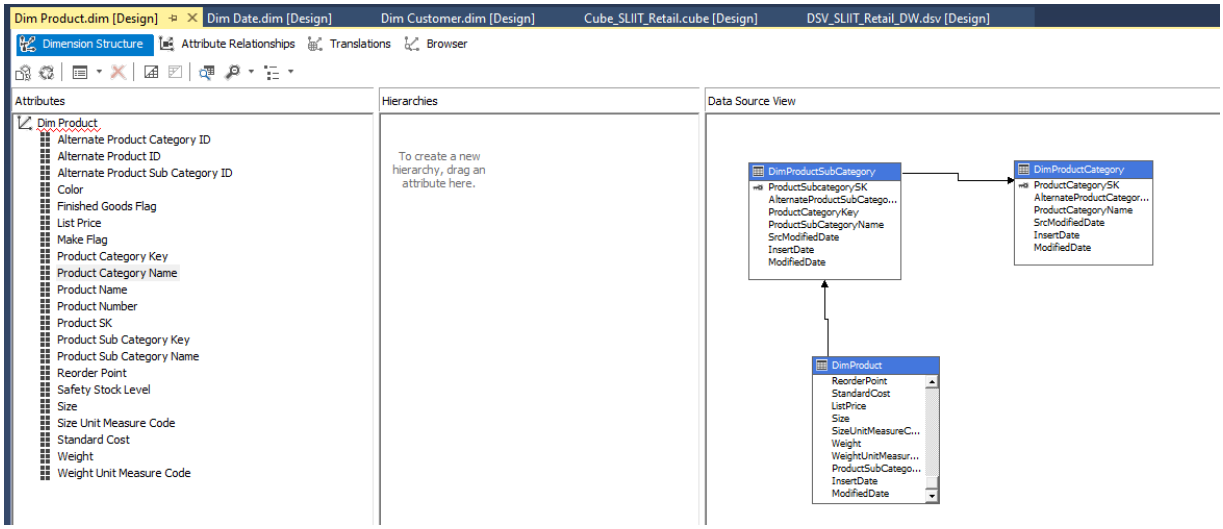
9. In the ***Dimensions*** section, expand '**Dim Customer**' and then '**Attributes**'. Note that only the '**Customer SK**' is listed as an attribute.



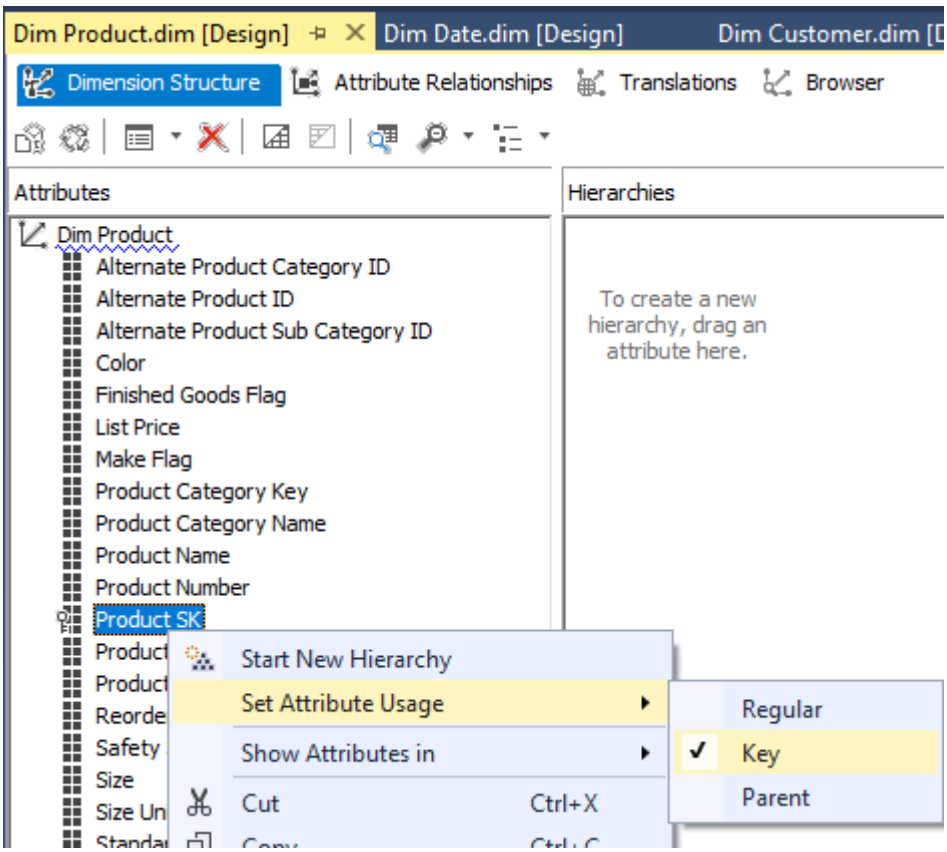
10. To add rest of the attributes, click on the '**Edit Dim Customer**' link, which will open dimension editor window for '**Dim Customer**'



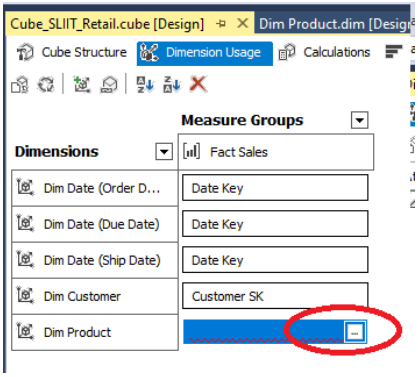
11. Select all the fields (except for '**CustomerSK**') in '**DimCustomer**' tables visible in ***Data Source View*** pane on right, drag and drop them on the ***Attribute*** pane on the left side.
12. Ensure to save the project.
13. Now, go back to '**Cube_SLIIT_Retail.cube**' design view (click on this name under ***Cubes*** in ***Solution Explorer*** is it is not already open), expand '**Order Date**' in the ***Dimensions*** section and click on '**Edit Dim Date**' and add all the attributes (except for '**DateKey**') by following the same steps you follow to add attributes to '**Dim Customer**'.
14. Follow same steps to add all the attributes of '**DimProduct**'. Note that when you try to add attributes related to 'Product', it includes product sub category and category attributes too since they are normalized tables under same subject area; "Product" (in reality fields such as 'SrcModifiedDate', 'InsertDate' and 'ModifiedDate' makes no meaning for an analysis. Thus, you may avoid those attributes).



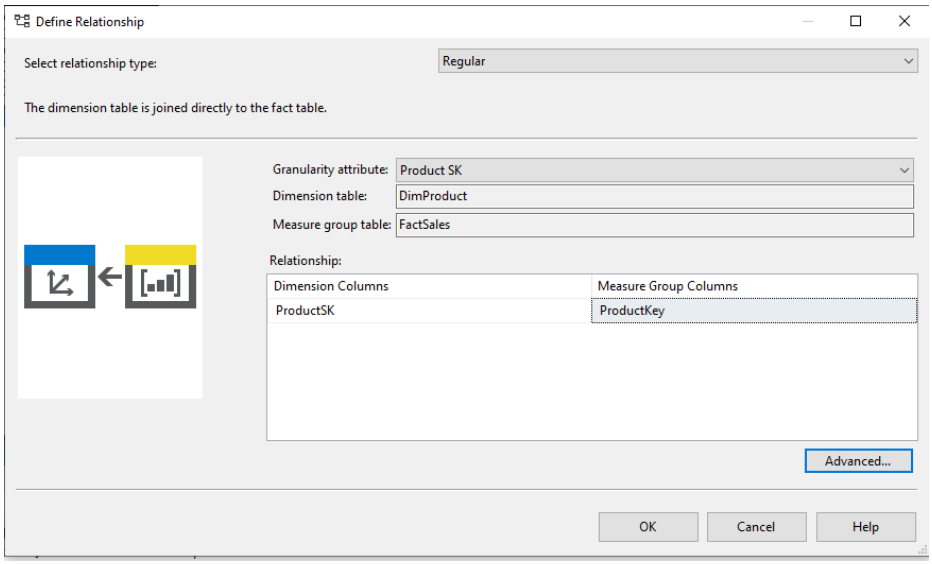
15. IMPORTANT: For '**Dim Product**', a primary key may not be set, if it is not automatically picked. If so, in the '**Dim Product**' design window ('**Dim Product.dim [Design]**'), under ***Dimenstion Structure*** tab, right click on '**Product SK**' attribute, select ***Set Attribute Usage*** → ***Key***.



16. Once you set the key for the product dimension, we need to define how the product is connected to fact table. To do this, go to '**Cube_SLIIT_Retail**' design window ('**Cube_SLIIT_Retail.cube [Design]**'), under ***Dimension Usage*** tab, for '**Dim Product**' under Dimensions column, you might see a value is not set. If so, click on the button available to set the value.



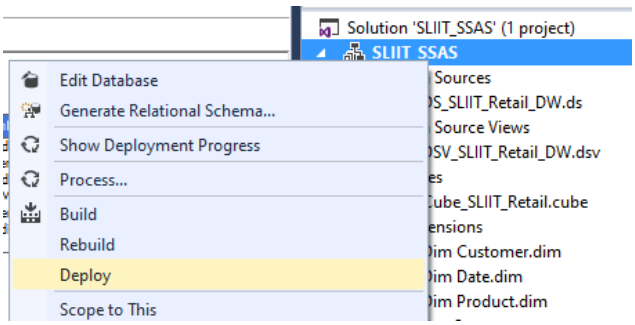
17. ***Define Relationship*** window will pop-up. Define the relationship as given in the image below. Dimension column "**ProductSK**" should be mapped with measure group column '**ProductKey**'.



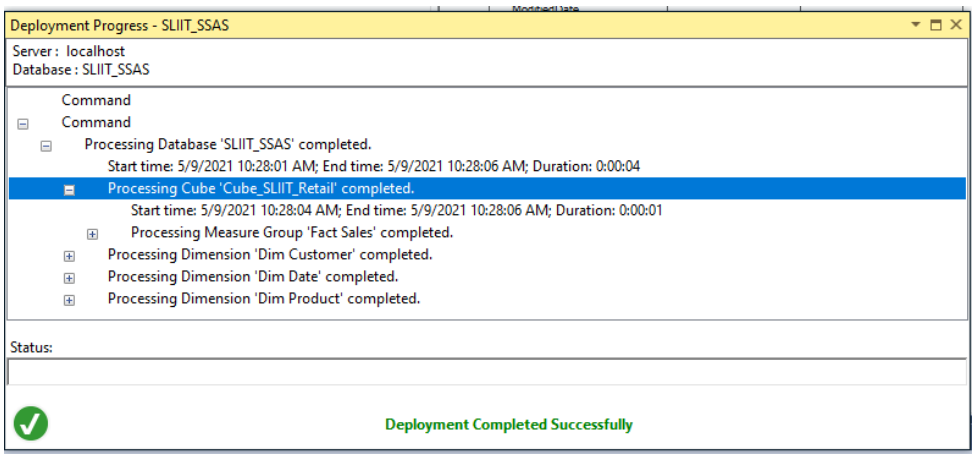
7. Deploy the Cube

The Cube must be deployed in order to be used for analysis. Having proper connection details provided will take effect in this stage when you try to deploy. Once deployed, SSAS Cube will be available for analysis under SSAS databases accessible via SSMS.

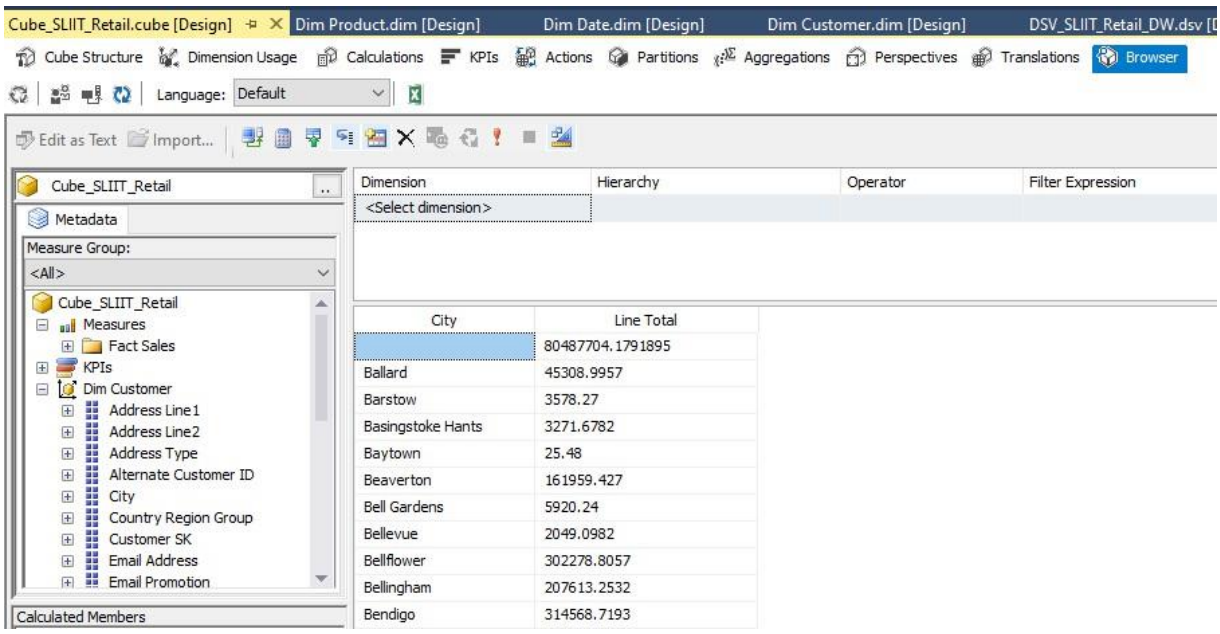
1. Right click on the project name, '**SLIIT_SSAS**' in solution explorer, and click on ***Deploy***.



You will see a pop-up window displaying the progress of the deployment.

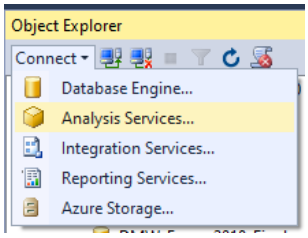


Once successfully deployed, ***Browser*** tab under '**Cube_SLIIT_Retail**' design window ('**Cube_SLIIT_Retail.cube [Design]**'), will have the attributes of the model on the left hand side, where you can drag and drop the into the design area on the right hand side and do

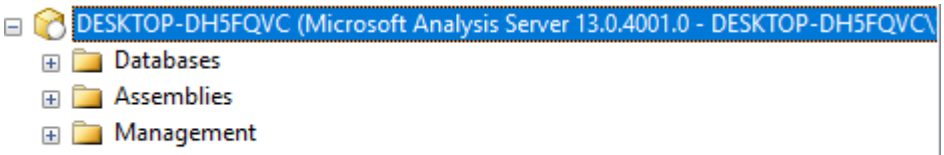


some test analysis.

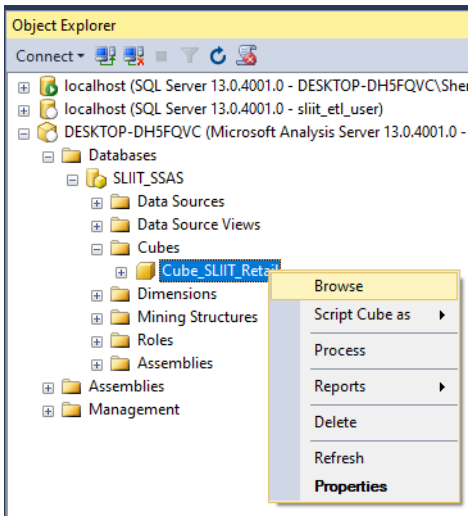
2. To check the deployment in SSMS, open SQL Server Management Studio, click on ***Connect*** and select ***Analysis Services…***



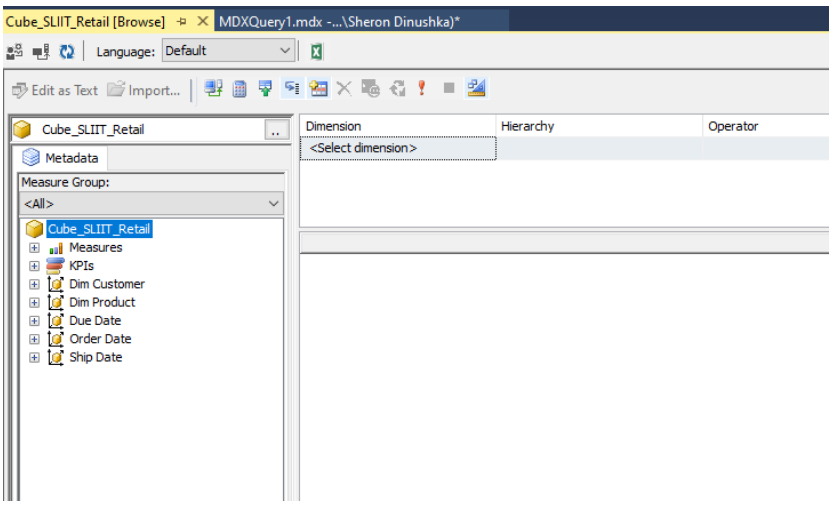
3. Provide credentials and log in. You should be able to see something similar to the below image once you logged in to Analysis Services.



4. You can expand ***Databases***, '**SLIIT_SSAS**', and right click on '**Cube_SLIIT_Retail**' and select ***Browse***.



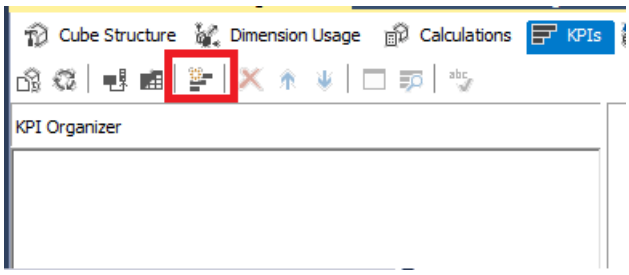
5. Same browser window you saw in Data Tools in '**Cube_SLIIT_Retail**' design window ('**Cube_SLIIT_Retail.cube [Design]**') should appear here allowing you to analyse data. Optionally you can write MDX, DAX queries.



8. Creating a KPI

KPI's are created based on the business requirements. KPIs depend on what the organization want to monitor and measure.

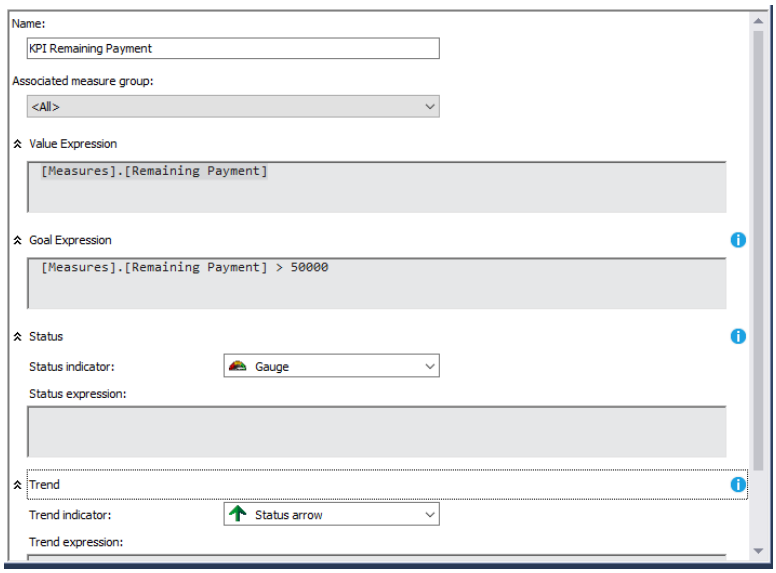
1. In order to create a sample KPI definition, go back to Data Tools, locate and go to ***KPIs*** tab in '**Cube_SLIIT_Retail**' design window ('**Cube_SLIIT_Retail.cube [Design]**'). If the design window is not visible you can double click on '**Cube_SLIIT_Retail.cube**' to open the design window.
2. In the ***KPIs*** tab, above ***KPI Organizer*** panel, locate and click on New ***KPI*** button. Alternatively you right click on ***KPI Organizer*** panel area and select ***New KPI***.



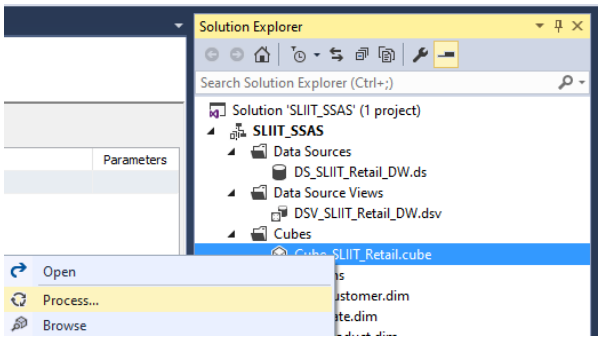
3. Name the KPI as '**KPI Remaining Payment**', select '**Fact Sales**' as the ***Associated measure group***.
4. In the ***Measure Group*** on the lower left-hand side panel, expand ***Measures*** and then expand '**Fact Sales**'. Drag and drop '**Remaining Payment**' attribute to ***Value Expression*** area.
5. Drag and drop '**Remaining Payment**' attribute to ***Global Expression*** area and modify the expression as follows:

```
[Measures].[Remaining Payment] > 50000
```

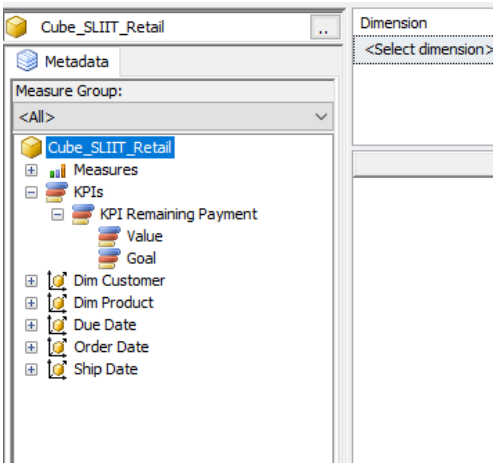
Goal Expression is the KPI!



6. Save all the changes.
7. Additionally, research on other options available under KPI feature in SSAS.
8. Now, if you go to '**Cube_SLIIT_Retail**' design window ('**Cube_SLIIT_Retail.cube [Design]**'), and ***Browser*** tab, and try to expand ***KPIs*** node, you will notice that the newly created ***KPI*** is not available (you will notice the same behavior in SSMS also). In order to update the cube, right click on the cube name in the Solution Explorer; '**Cube_SLIIT_Retail.cube**' and click ***Process*** (Same thing can be done in SSMS also).



9. Now you can include the KPI you created in your analysis.



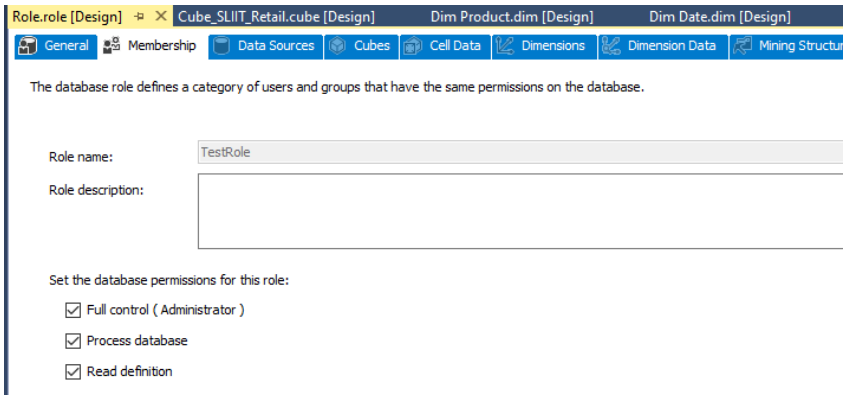
10. Do some research on creating hierarchies to dimensions, adding new measures to the cube, creating Business Intelligence, etc…

9. Creating a Role

Purpose of user roles is to provide control access and permission on who can do what.

1. To add a role, right click on ***Roles*** in ***Solution Explorer*** and select ***New Role***.

2. In the role design page (***Role.role [Design]***), ***General*** tab, provide a role description and select ***Full control (Administrator)***, to give full control to the new role. You can rename the role name in the properties windows.

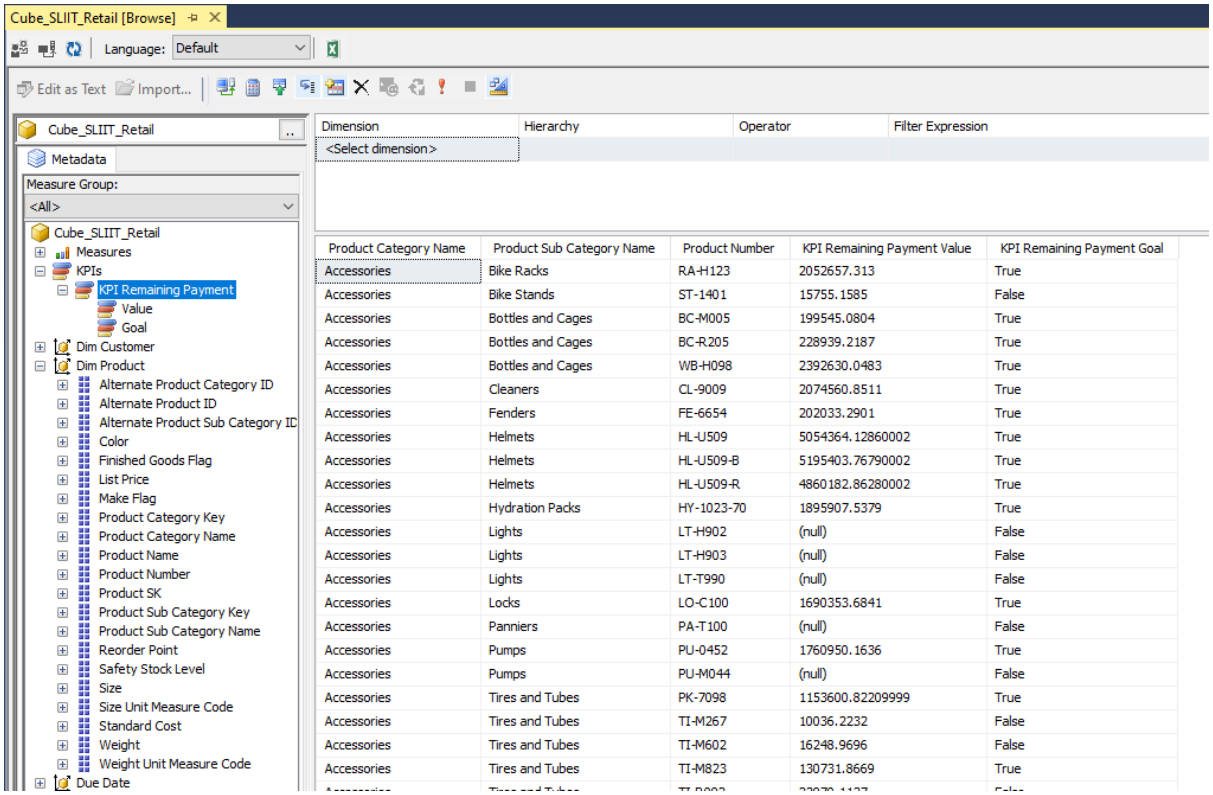


3. Now go to the ***Membership*** tab and click on the ***Add…*** button. You can add a user to the newly created Role.
4. Save all the changes.

10. Browsing Cube Data

You have already tried this. As we already know general browsing (analysis) can be done via the development tool; Data Tools or in SSMS.

A sample is given below:



Product Category Name	Product Sub Category Name	Product Number	KPI Remaining Payment Value	KPI Remaining Payment Goal
Accessories	Bike Racks	RA-H123	2052657.313	True
Accessories	Bike Stands	ST-1401	15755.1585	False
Accessories	Bottles and Cages	BC-M005	199545.0804	True
Accessories	Bottles and Cages	BC-R205	228939.2187	True
Accessories	Bottles and Cages	WB-H098	2392630.0483	True
Accessories	Cleaners	CL-9009	2074560.8511	True
Accessories	Fenders	FE-6654	202033.2901	True
Accessories	Helmets	HL-U509	5054364.12860002	True
Accessories	Helmets	HL-U509-B	5195403.76790002	True
Accessories	Helmets	HL-U509-R	4860182.86280002	True
Accessories	Hydration Packs	HY-1023-70	1895907.5379	True
Accessories	Lights	LT-H902	(null)	False
Accessories	Lights	LT-H903	(null)	False
Accessories	Lights	LT-T990	(null)	False
Accessories	Locks	LO-C100	1690353.6841	True
Accessories	Panniers	PA-T100	(null)	False
Accessories	Pumps	PU-0452	1760950.1636	True
Accessories	Pumps	PU-M044	(null)	False
Accessories	Tires and Tubes	PK-7098	1153600.82209999	True
Accessories	Tires and Tubes	TI-M267	10036.2232	False
Accessories	Tires and Tubes	TI-M602	16248.9696	False
Accessories	Tires and Tubes	TI-M823	130731.8669	True